SPEAKDECK

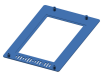
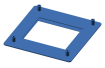
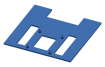
Build instructions



A Spotify player and OctoPrint host with the size of one Denon SC-N10 speaker.

153 mm wide. 233 mm high. 196 mm deep.

Printed parts

	Part	File	Quantity	Size on the bed (mm)	Filament	Length
	Body	`stl/body.stl`	1	153 × 264 × 193	558 g	187.3 m
	Front frame	`stl/frame.stl`	1	141 × 225 × 10	40 g	13.5 m
	Top frame	`stl/top_frame.stl`	1	141 × 143 × 12	35 g	11.8 m
	Rear panel	`stl/rear.stl`	1	146 × 222 × 3	104 g	35.0 m
	Sled	`stl/sled.stl`	1	146 × 146 × 10	53 g	17.6 m
	Drive tray	`stl/tray.stl`	1	146 × 176 × 4	72 g	24.1 m
	Cap	`stl/cap.stl`	8	13 × 13 × 7	6 g	2.1 m
	Inky pad (TPU)	`stl/pad.stl`	1	90 × 76 × 2	14 g	4.8 m
	Total				**883 g**	**296 m**

The filament values are estimates for PLA, 1.75 mm filament, 1.2 mm walls, and 15 % infill. The values for a part with a quantity include all pieces. A slicer gives the correct values for your settings.

- The STL files are in the `case/stl/` directory. Each file has the correct position for the print bed.
- Print the body with the front face on the bed. Use supports for the frame indent and the top opening.
- Print the two frames in a dark color.
- Print the pad in TPU, or use the file as a pattern and cut the pad from 2 mm rubber or foam.

Hardware

16	M3 heat-set insert, 4 mm long
16	M3 × 6 bolt
4	M3 × 8 countersunk bolt
4	M3 × 16 bolt and M3 nut
4	Speaker binding post (2 red, 2 black)
1	DC panel jack, 5.5 × 2.1 mm
4	Rubber foot, self-adhesive

Electronics

1	Raspberry Pi 5 + adapter + HiFiBerry Amp2, as one stack
1	Inky wHAT with a 40-pin ribbon cable, 200 mm
1	Adafruit 7 inch HDMI display with touch
1	2.5 inch SATA drive with a USB adapter board
1	30 mm fan
1	Micro HDMI to HDMI cable, right-angle HDMI plug
1	USB-A to micro USB cable, right-angle micro USB plug
1	Short USB cable for the drive

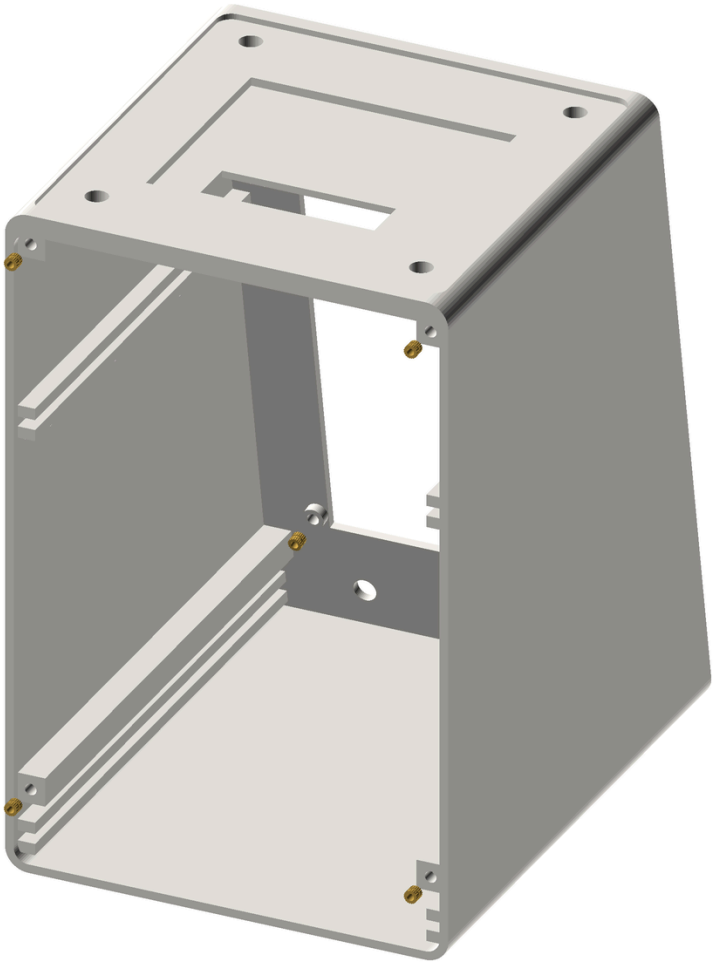
Tools: soldering iron with an insert tip, hex keys for M3, small flat screwdriver, calipers.

Measure first. Compare your Inky wHAT (thickness) and the height of your Pi stack with the values in `speakdeck.scad`. Correct the values, then run `build/make.sh`.

1 Inserts in the body

8 × M3 insert

Printed parts: `stl/body.stl`



1. Push 4 inserts into the 4 bosses around the display opening. Work through the rear opening.
2. Push 4 inserts into the ends of the corner posts.
3. Let the body become cool.

2 Inserts in the frames

8 × M3 insert

Printed parts: `stl/frame.stl`, `stl/top_frame.stl`



1. Push 4 inserts into the 4 bosses on the rear face of the front frame.
2. Push 4 inserts into the 4 bosses on the lower face of the top frame.

3 Front frame on the body

4 × Cap

4 × M3 × 6

Printed parts: `stl/body.stl`, `stl/frame.stl`, `stl/cap.stl`

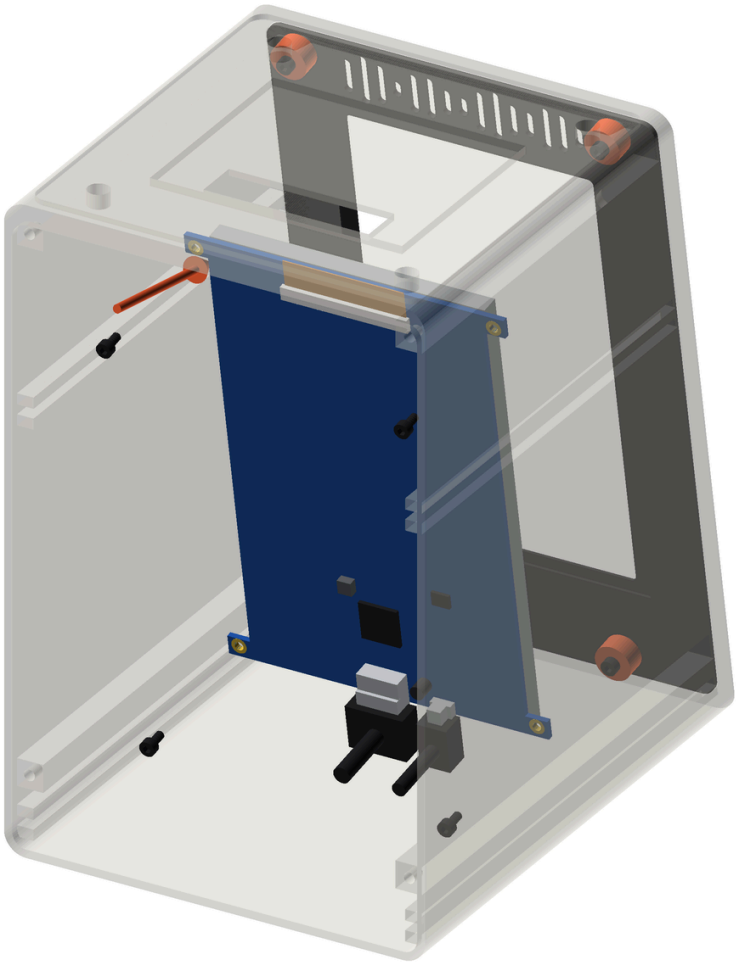


1. Put the front frame into the indent in the front of the body. The pattern is at the top.
2. The 4 bosses of the frame go through the 4 holes of the body.
3. Put a cap on each boss from the inner side.
4. Install a bolt through each cap.

4 Display into the body

4 × M3 × 6

Printed parts: `stl/body.stl`



1. Put the body on a soft surface. The front points down.
2. Connect the right-angle HDMI plug and the right-angle USB plug to the display.
3. Put the display through the rear opening. The glass points to the front, and the sockets point to the bottom.
4. Put the LCD module into the opening in the front. The board sits on the 4 bosses.
5. Install 4 bolts through the 4 holes of the display board.

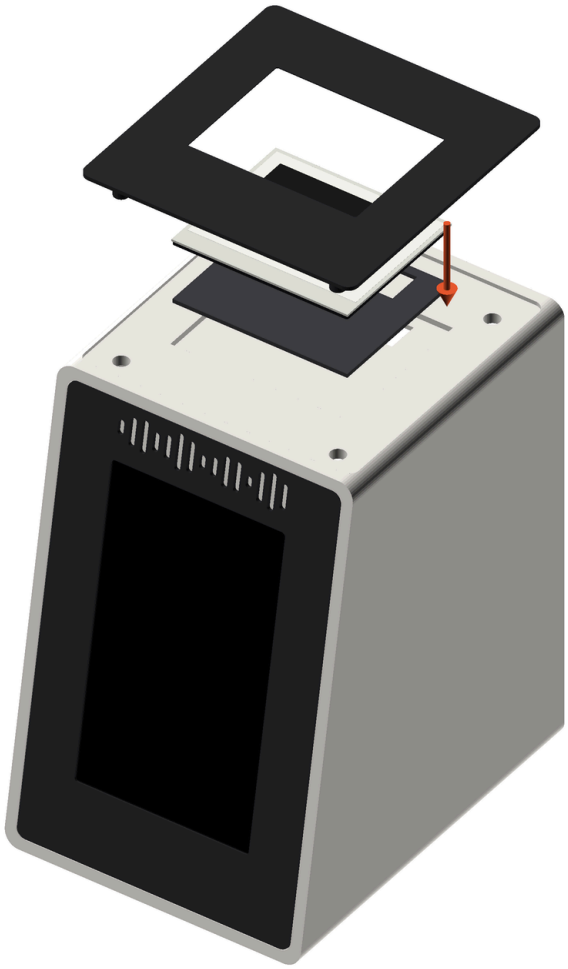
The display glass is now 1 mm behind the front of the frame.

Caution. Tighten the bolts lightly. Too much force can bend the board.

5 Inky wHAT in the top

1 × Pad | **4 ×** Cap | **4 ×** M3 × 6

Printed parts: `stl/top_frame.stl`, `stl/pad.stl`, `stl/cap.stl`



1. Put the pad into the cavity in the top of the body.
2. Connect the ribbon cable to the header of the Inky wHAT.
3. Put the ribbon cable through the opening in the cavity.
4. Put the Inky wHAT on the pad. The header goes into the opening.
5. Put the top frame into the indent. The frame pushes the Inky wHAT down on the pad.
6. Put a cap on each boss from the inner side, and install a bolt through each cap.

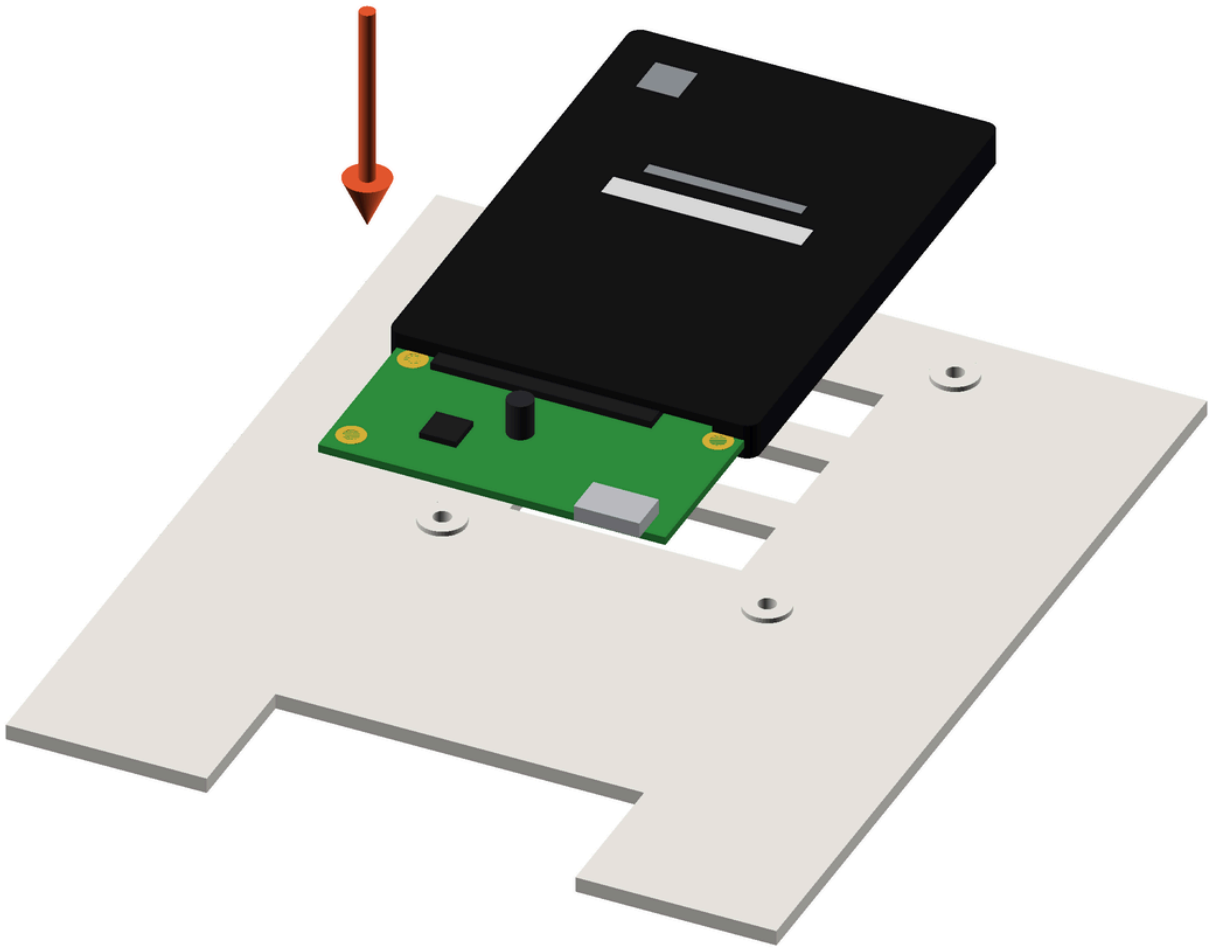
The e-paper is now 0.6 mm below the top of the frame.

Caution. Do not push on the e-paper. The glass is thin.

6 Drive on the tray

4 × M3 × 6

Printed parts: `stl/tray.stl`

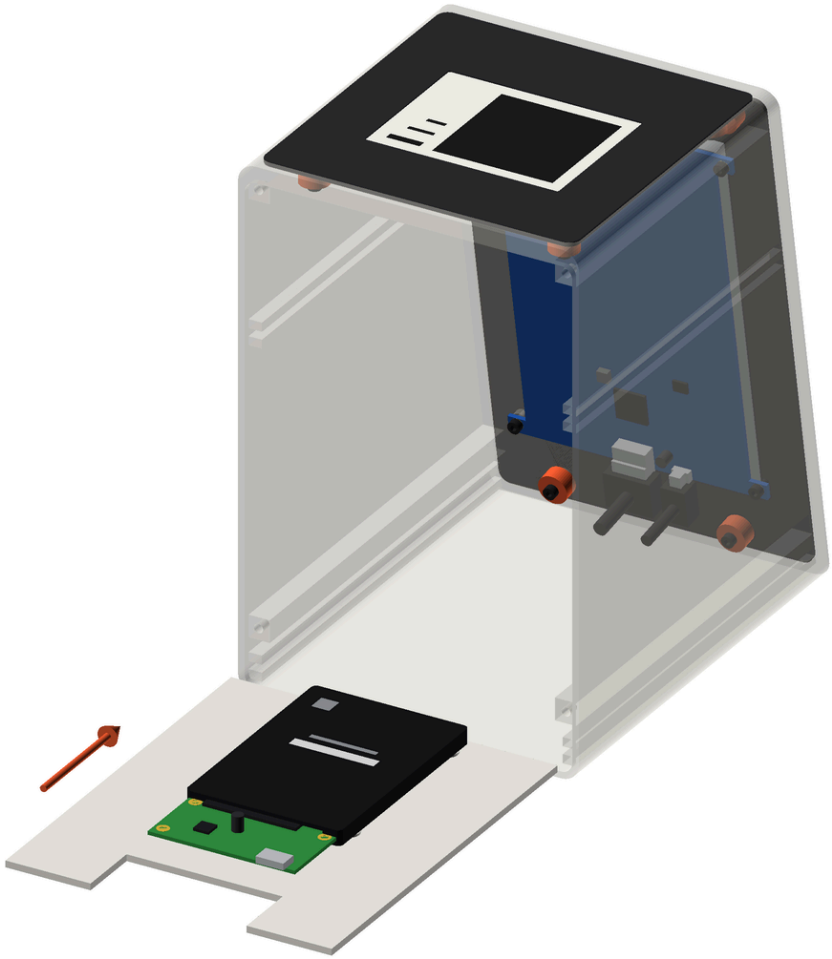


1. Put the drive on the 4 pads of the tray. The label points down, and the adapter board points to the cable opening at the rear.
2. Install 4 bolts from below, through the tray, into the 4 holes in the bottom of the drive.

Caution. Use M3 × 6 bolts or shorter. A longer bolt can damage the drive.

7 Drive tray into the body

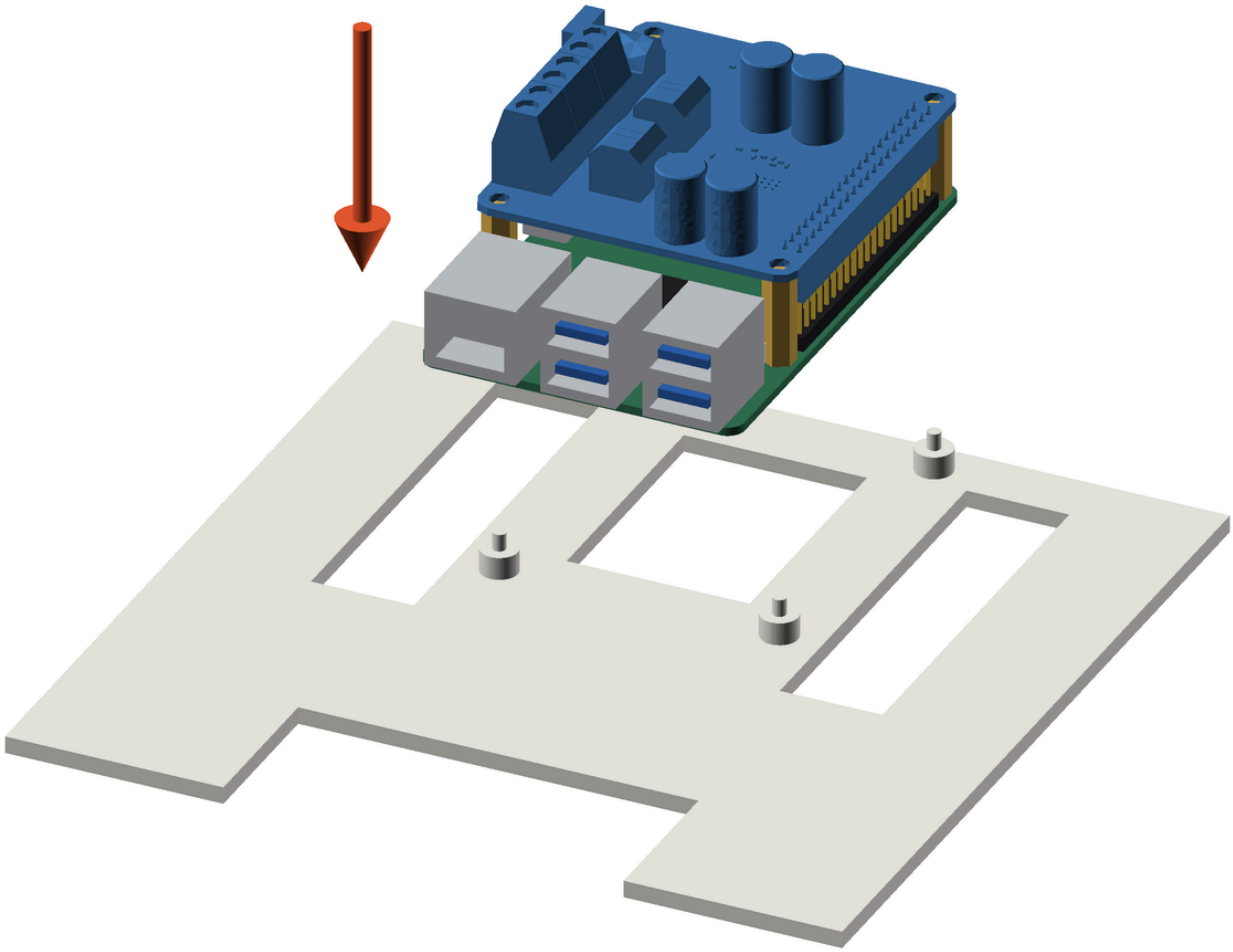
Printed parts: `stl/tray.stl`



1. Put the 2 display cables below the tray position.
2. Put the tray into the lower grooves.
3. Push the tray in fully.

8 Pi stack on the sled

Printed parts: `stl/sled.stl`

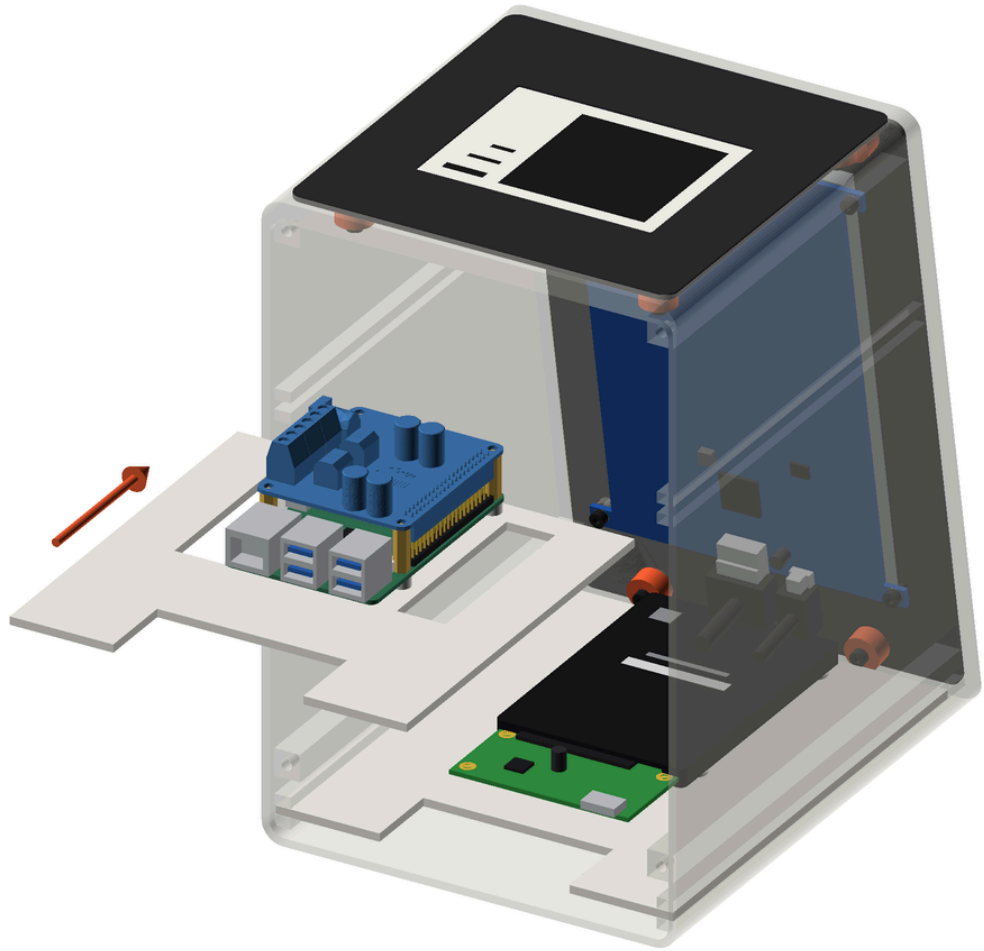


1. Put the Pi stack on the 4 pins of the sled.
2. Make sure that the USB ports point to the cable opening at the rear.

The pins and the weight of the stack hold it in position. Keep the case upright.

9 Sled into the body

Printed parts: `stl/sled.stl`



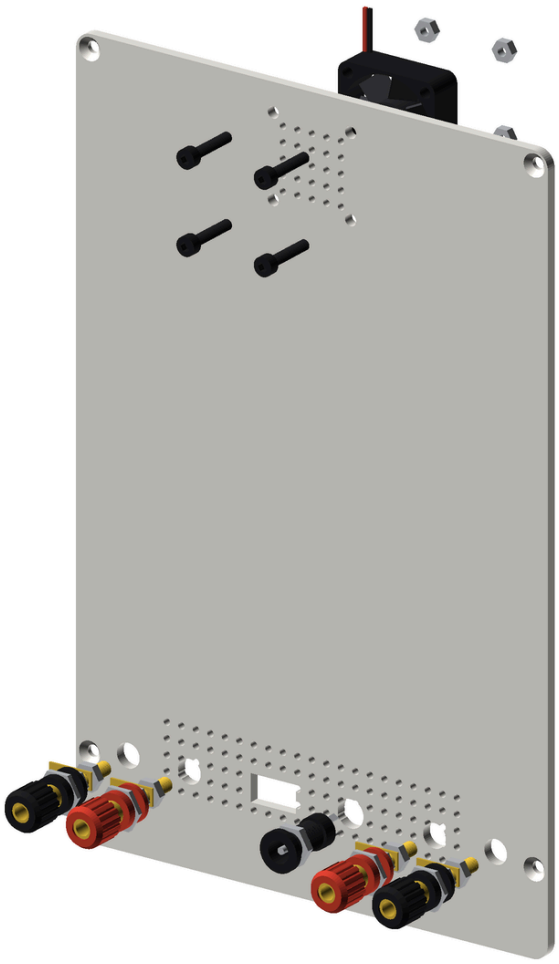
1. Put the sled into the upper grooves. Push it in halfway.
2. Connect the ribbon cable of the Inky wHAT to the header on top of the stack.
3. Push the sled in fully.

Caution. Make sure that pin 1 of the ribbon cable is on pin 1 of the header.

10 Rear panel

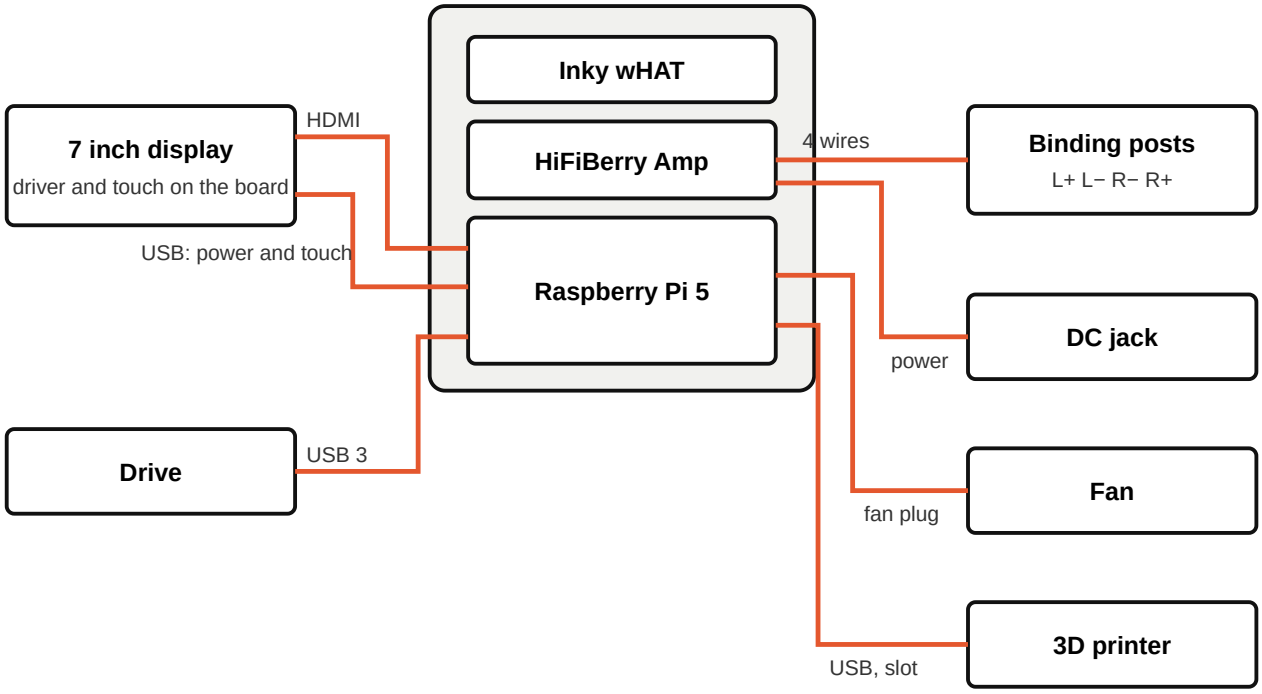
4 × M3 × 16 + nut | **4 ×** Binding post | **1 ×** DC jack

Printed parts: `stl/rear.stl`



1. Attach the fan to the inner side, behind the round hole pattern. The label of the fan points to the panel, so the air goes out of the case.
2. Install the 4 binding posts. Put red on the inner holes and black on the outer holes.
3. Install the DC jack in the round hole near the center.

11 Cables



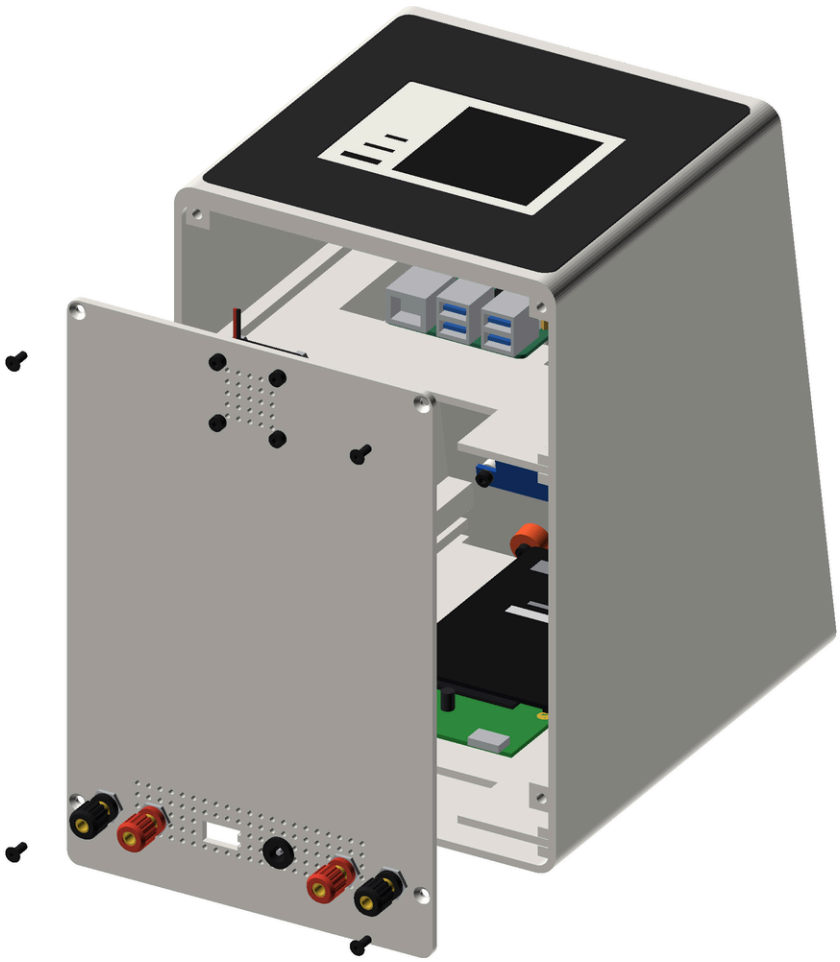
1. Connect the HDMI cable of the display to a micro HDMI port of the Pi.
2. Connect the USB cable of the display to a USB port of the Pi.
3. Connect the drive to a blue USB port of the Pi.
4. Connect the binding posts to the speaker terminals of the Amp. Keep the polarity correct.
5. Connect the DC jack to the power input of the Amp.
6. Connect the fan to the fan connector of the Pi.
7. Put the USB cable of the printer through the slot in the rear panel.

Warning. The Amp supplies power to the Pi. Do not connect a USB-C power supply at the same time.

12 Close the case

4 × M3 × 8 countersunk

Printed parts: `stl/rear.stl`



1. Put the rear panel into the rear of the body.
2. Make sure that no cable is between the panel and the trays.
3. Install the 4 bolts into the corner posts.

13 Complete

4 × Rubber foot



1. Attach the 4 rubber feet to the bottom.
2. Put the unit between the speakers.
3. Connect the speaker cables to the binding posts.
4. Connect the 12 to 24 V power supply.